

# Xiang Li 李翔

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## About me

I am a first-year Ph.D. student in a joint training programme between **BNU–HKBU United International College** and **Great Bay University**. I received my M.Sc. in Biological Engineering from the **University of Saskatchewan** (2026), where I worked with Prof. Yen-Han Lin on machine learning approaches for redox potential–controlled very-high-gravity ethanol fermentation, and my B.Eng. in Biological Engineering from **Northwest A&F University** (2024).

My research sits where **fermentation bioprocess engineering** meets **machine learning**. I build soft sensors and kinetic models that turn cheap, continuous online signals — oxidation–reduction potential (ORP) in particular — into quantitative, forward-looking estimates of metabolic activity, so that a fermentation can be steered before it fails rather than analysed after it ends.

**Position:** Ph.D. Student, Year 1  
**Programme:** BNU–HKBU UIC × Great Bay University  
**M.Sc.:** University of Saskatchewan, 2026 (GPA 4.0/4.0)  
**B.Eng.:** Northwest A&F University, 2024

I am open to research collaborations and discussions on machine learning for bioprocess monitoring, kinetic modelling, and fermentation process control.

## Research interests 研究方向

Fermentation process control;    Kinetic modelling & parameter inference;    Deep learning for bioprocesses (AI4Bio);  
Soft sensing & state estimation;    Bayesian uncertainty quantification;    Bioenergy & cleaner biomanufacturing

## Publications 学术成果

### Bayesian inference on fermentation kinetics: Comparative analysis with frequentist approach

PUBLISHED

Xiang Li, Yen-Han Lin

*Bioresource Technology*, 2026, 444, 134012. IF 9.0, JCR Q1.

DOI → (<https://doi.org/10.1016/j.biortech.2026.134012>)    ScienceDirect → (<https://www.sciencedirect.com/science/article/pii/S0960852426000933>)

Enhancing cleaner biomanufacturing of ethanol: A temporal deep learning soft sensor for ORP-driven metabolic forecasting in very-high-gravity fermentation UNDER REVIEW

Xiang Li, Yen-Han Lin

Journal of Cleaner Production — under review, 2026.

Modified Hyperbolic Secant (MHS) Models for Fermentation Kinetics and Uncertainty POSTER

Xiang Li, Yen-Han Lin

Poster, YABEC 2026 (Asian Young Biotechnologists Network), University of Saskatchewan.

CV 简历

EDUCATION

Ph.D. Student, Biological Engineering 2026 – present

BNU–HKBU United International College & Great Bay University, Guangdong, China

Joint training programme. Year 1.

M.Sc., Biological Engineering 2025 – 2026

University of Saskatchewan, Saskatoon, Canada

GPA 4.0/4.0. Supervisor: Prof. Yen-Han Lin. Thesis: *Machine learning approaches for redox potential-controlled very high gravity ethanol fermentation.*

B.Eng., Biological Engineering 2020 – 2024

Northwest A&F University, Yangling, China

GPA 3.71/4.0, rank 3/63. Thesis: *Deep learning-based image processing of maize allele-specific chromatin interactions and regulation of imprinted genes.*

RESEARCH EXPERIENCE

Deep learning monitoring and forecasting system for ethanol fermentation Jun. 2025 – present

- Fused sparse offline measurements with dense online ORP signals to build a high-resolution training set; developed a real-time state-estimation model (ORPSN, RNN) and a multi-horizon forecasting model (ORPSC, LSTM encoder–decoder) for metabolic dynamics.
- Outcome:** Achieved high-accuracy soft sensing of ethanol production and glucose consumption rates; maintained accuracy over an 8-hour forecast horizon, providing a basis for proactive feed-forward control.

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| <b>Bayesian inference for fermentation kinetics and uncertainty analysis</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Jan. 2025 – Dec. 2025 |
| <ul style="list-style-type: none"> <li>Built Bayesian hierarchical kinetic models for multiple fermentation systems with algorithmic and literature-informed priors; systematically compared Bayesian and frequentist inference for parameter identifiability and predictive stability under sparse data.</li> <li><b>Outcome:</b> Resolved non-convergence of conventional methods in small-sample settings, giving robust kinetic parameter estimates with explicitly quantified uncertainty and improved biological interpretability.</li> </ul> |                       |

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| <b>Allele-specific chromatin interactions and imprinted gene regulation in maize endosperm</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Mar. 2023 – May 2024 |
| <ul style="list-style-type: none"> <li>Applied autoencoders and U-Net to denoise and enhance the resolution of Hi-C data from reciprocal crosses of maize inbred lines B73 and Mo17; constructed allele-specific chromatin interaction maps; partitioned A/B compartments and identified TADs genome-wide; analysed compartment distributions around imprinted genes.</li> <li><b>Outcome:</b> Built an allele-resolved 3D chromatin architecture of maize endosperm and showed that the compartmental distribution of imprinted genes has no clear parent-of-origin dependence, indicating that chromatin reorganisation at the A/B compartment level is not the principal mechanism of imprinting regulation there.</li> </ul> |                      |

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| <b>Metagenomics and culturomics for tobacco biological control</b>                                                                                                                                                                                                                                                                                                                | Nov. 2021 – Nov. 2022 |
| <ul style="list-style-type: none"> <li>Screened and validated antagonistic strains and assembled consortia; designed and ran field trials; performed 16S rRNA amplicon analysis of the tobacco rhizosphere microbiome.</li> <li><b>Outcome:</b> Established a biocontrol strain library with good field efficacy and developed an effective composite microbial agent.</li> </ul> |                       |

RESEARCH VISITS & INDUSTRY

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|-------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Visiting Student</b>                                                                               | Jan. 2026 – present   |
| Shanghai Jiao Tong University, School of Life Sciences and Biotechnology                              |                       |
| Project: deep learning–based metabolic modelling of filamentous fungi and its industrial application. |                       |
| <b>Sales Engineer</b>                                                                                 | Sep. 2024 – Dec. 2024 |
| Servicebio Technology Co., Ltd., Zhengzhou, China                                                     |                       |
| Technical support and product application for life-science research reagents.                         |                       |

HONOURS & AWARDS

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|-----------------------------------------------------------------------------------------------|-----------|
| University of Saskatchewan CGPS 75th Recruitment Scholarship                                  | Jan. 2025 |
| National Scholarship for Encouragement (国家励志奖学金), Northwest A&F University                    | Nov. 2021 |
| First Prize, National Undergraduate Life Sciences Competition (Innovation & Entrepreneurship) | Nov. 2021 |

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| Meritorious Winner (top 8%), Mathematical Contest in Modeling, COMAP, USA            | Apr. 2021 |
| First-Class Academic Scholarship, College of Life Sciences, Northwest A&F University | Dec. 2020 |

SKILLS

Python; PyTorch; TensorFlow / Keras; NumPy & pandas; Bayesian inference (MCMC); Kinetic modelling;  
Bioreactor operation & ORP control; HPLC analysis; R; Git